

Vertex fault tolerance for multiple spanning paths in hypercube *

Chun-Nan Hung and Guan-Yu Shi

Department of Computer Science and Information Engineering

Da-Yet University, Changhua, Taiwan 51505, R.O.C

spring@mail.dyu.edu.tw, r9406017@mail.dyu.edu.tw

Abstract

Let $\{s_i, t_i\}_{F_w, K_b}^{F_b, K_b}$ be a family of $G = (V_b \cup V_w, E)$ where $K_b \subset V_b$ and $K_w \subset V_w$ such that $\{s_i, t_i | 1 \leq i \leq (|K_b| + |K_w|)/2\}$ is the set of fault-free vertices, $F_b \subset V_b$ and $F_w \subset V_w$ are sets of faulty vertices. The family $\{s_i, t_i\}_{F_w, K_w}^{F_b, K_b}$ is balanced if $|K_w| + 2|F_w| = |K_b| + 2|F_b|$. The family $\{s_i, t_i\}_{F_w, K_w}^{F_b, K_b}$ is connectable if there exist $(|K_b| + |K_w|)/2$ spanning paths $P(s_i, t_i)$, for $1 \leq i \leq (|K_b| + |K_w|)/2$, in $G - F_b - F_w$. In this paper, we will show that every balanced family $\{s_i, t_i\}_{F_w, K_w}^{F_b, K_b}$ of hypercube Q_n is connectable if $|F_b| + |F_w| + |K_b| + |K_w| \leq n$, $4|F_b| + 2|K_b| \leq n + 1$ and $4|F_w| + 2|K_w| \leq n + 1$, for $n \geq 3$.

1 Introduction

The hypercube network is one of the most popular and efficient topological structures of interconnection networks. This is partially due to its many excellent features, such as regularity, symmetry, low diameter and degree, high connectivity, simple routing algorithms, and so on. Linear array and rings are two of the most fundamental interconnection networks. They are suitable for designing simply algorithms with low costs. Component failures are unavoidable in a large parallel systems. Thus, fault tolerance is an important research issue for interconnection networks.

An interconnection network is usually represented by a graph where vertices represent processors and edges represent links between processors. Let $G = (V_b \cup V_w, E)$ be a bipartite graph where V_b and V_w are two disjoint vertex sets such that each edge of E consists of one vertex from each set. A bipartite graph $G = (V_b \cup V_w, E)$ is *Hamiltonian laceable* if there exists a *Hamiltonian path*

between x, y for any $x \in V_b, y \in V_w$. The graph $G = (V_b \cup V_w, E)$ is *hyper-Hamiltonian laceable* if $\forall v \in V_b(\text{resp. } V_w)$, there exists a Hamiltonian path of $G - \{v\}$ between each pair of vertices of $V_w(\text{resp. } V_b)$. In [10], Tsai et al. proved that $Q_n - F_e$ is Hamiltonian laceable for $F_e \subset E(Q_n)$ and $|F_e| \leq n-2$, and $Q_n - F_e$ is hyper-Hamiltonian laceable for $|F_e| \leq n-3$.

There is no literature about general vertex fault tolerant Hamiltonian properties of hypercube for the number with faulty vertices more than one. Some literature concerns embedding fault-free cycles or paths for hypercube with faulty vertices. The upper bound of longest fault-free cycle of $Q_n(V_b \cup V_w, E) - F_v$ is $2^n - 2f$ where F_v is the faulty set of vertices of Q_n and $f = \max\{|F_v \cap V_w|, |F_v \cap V_b|\}$. In [2, 4, 7, 9], the authors showed that a fault-free cycle of length $2^n - 2f_v$ can be constructed with f_v faulty vertices. Fu [3] also showed that there exists a fault-free path of length $2^n - 2f_v - 1$ (resp. $2^n - 2f_v - 2$) between two arbitrary vertices of odd (resp. even) distance. These lengths are optimal when all the faulty vertices are in the same partite set. However, there exist longer cycles and paths when both partite vertex sets contain some faulty vertices. In [5], Hung et al. showed that there exists a fault-free cycle of length $2^n - 2|F_v| + 2$ in $Q_n - F_v$ where F_v is the set of faulty vertices with at least two vertices of odd distance. Su. et al. furthermore constructed the fault-free cycle with length $2^n - 2|F_v| + 4$ in $Q_n - F_v$ where $F_v \subset (V_b \cup V_w)$ with $|F_v \cap V_b| \geq 2$ and $|F_v \cap V_w| \geq 2$ in [8].

In [1], Caha et al. proposed the multiple spanning paths problem for hypercube. Let s_i, t_i , for $1 \leq i \leq k$, be vertices of Q_n . The $\{s_i, t_i\}_{i=1}^k$ is a *connectable family* if there exists k spanning paths of Q_n between s_i and t_i , for $1 \leq i \leq k$. The $\{s_i, t_i\}_{i=1}^k$ is *balanced* if it has the same number of vertices in each partite set. Caha showed that every balanced family $\{s_i, t_i\}_{i=1}^n$ is connectable in

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Q_{2n} if the distance of every pair s_i, t_i is odd. Caha also showed that every balanced family $\{s_i, t_i\}_{i=1}^n$ is connectable in Q_{6n} .

In this paper, we investigate the vertex fault tolerance for multiple spanning paths in hypercube. Let $\{s_i, t_i\}_{F_b, K_b}^{F_b, K_b}$ be a family of $G = (V_b \cup V_w, E)$ where $K_b \subset V_b$ and $K_w \subset V_w$ are sets of faulty vertices. The family $\{s_i, t_i\}_{F_b, K_b}^{F_b, K_b}$ is balanced if $|K_b| + 2|F_w| = |K_b| + 2|F_b|$. The family $\{s_i, t_i\}_{F_b, K_b}^{F_b, K_b}$ is connectable if there exist $(|K_b| + |K_w|)/2$ spanning paths $P(s_i, t_i)$, for $1 \leq i \leq (|K_b| + |K_w|)/2$, in $G - F_b - F_w$. In this paper, we will show that every balanced family $\{s_i, t_i\}_{F_b, K_b}^{F_b, K_b}$ of hypercube Q_n is connectable if $|F_b| + |F_w| + |K_b| + |K_w| \leq n$, $4|F_b| + 2|K_b| \leq n + 1$ and $4|F_w| + 2|K_w| \leq n + 1$, for $n \geq 3$. Thus, every balanced family $\{s_i, t_i\}_{F_b, K_b}^{F_b, K_b}$ is connectable in Q_{2n} . Applying this result, we can construct the fault-free cycle of $Q_n = (V_b \cup V_w, E) - F_v$ of length $2^n - 2f_v$ where $F_v \subset (V_b \cup V_w)$ and $f_v = \max\{|F_v \cap V_b|, |F_v \cap V_w|\} \leq \lceil \frac{n}{4} \rceil - 1$. We can also construct longer paths in Q_n with some faulty vertices.

The rest of this paper is organized as follows. In Section 2, we introduce some important definitions and notations. We show the vertex fault tolerance for multiple spanning paths in Section 3. Some conclusion is given in Section 4.

2 Definitions and Notations

An n -dimensional hypercube $Q_n(V_b \cup V_w, E)$ is a bipartite graph whose vertices are labeled by distinct n -bit binary strings. Two vertices are linked by an edge if and only if their labels differ exactly in one bit. An i -edge (x, y) is an edge that x and y differ in the i -th bit. The hypercube Q_n can be constructed recursively as $Q_n = Q_{n-1} \times K_2$. We can partition Q_n as two subgraphs Q_{n-1}^0 and Q_{n-1}^1 by choosing any one bit of binary string.

We call V_b black vertex set and V_w white vertex set. Let V_b^j and V_w^j the black and white vertex set of Q_{n-1}^j , for $j = 0, 1$. And let $V^j = V_b^j \cup V_w^j$ for $j = 0, 1$. Thus, $V_b = V_b^0 \cup V_b^1, V_w = V_w^0 \cup V_w^1, V = V_b \cup V_w = V^0 \cup V^1$.

Let F_b be the set of black faulty vertices of Q_n and F_w be the set of white faulty vertices of Q_n . Similarly, we also use F_b^j and F_w^j to denote the black and white faulty vertex set of Q_{n-1}^j , respec-

tively, for $j = 0, 1$. Thus, $F_b = F_b^0 \cup F_b^1, F_w = F_w^0 \cup F_w^1, F^0 = F_b^0 \cup F_w^0, F^1 = F_b^1 \cup F_w^1, F = F_b \cup F_w = F^0 \cup F^1$.

We define some functions for every vertex of V . Let $sub(v) = j$ if $v \in V(Q_{n-1}^j)$ for $j = 0, 1$. Let $color(v) = b$ if $v \in V_b$, $color(v) = w$ if $v \in V_w$. Let $\phi(v)$ be a vertex of V^1 for every $v \in V^0$ such that $(v, \phi(v)) \in E$. Let $\phi(A) = \{\phi(v) | v \in A\}$ for any vertex subset $A \subset V^0$.

Let K_b and K_w be the black and white fault-free vertex set. Let $K = K_b \cup K_w = \{s_i, t_i | 1 \leq i \leq \frac{|K|}{2}\}$. And let $K_b^j = K_b \cap V^j, K_w^j = K_w \cap V^j$, for $j = 0, 1$.

Moreover, let $K_{bw}^{01} = \{s_i, t_i | sub(s_i) \neq sub(t_i), color(s_i) \neq color(t_i) \text{ for } 1 \leq i \leq \frac{|K|}{2}\} \cap K_b^0$, let $K_{bb}^{01} = \{s_i, t_i | sub(s_i) \neq sub(t_i), color(s_i) = color(t_i) = b \text{ for } 1 \leq i \leq \frac{|K|}{2}\} \cap K_b^0$, $K_{ww}^{01} = \{s_i, t_i | sub(s_i) \neq sub(t_i), color(s_i) = color(t_i) = w \text{ for } 1 \leq i \leq \frac{|K|}{2}\} \cap K_w^0$, $K_{wb}^{01} = \{s_i, t_i | sub(s_i) \neq sub(t_i), color(s_i) \neq color(t_i) \text{ for } 1 \leq i \leq \frac{|K|}{2}\} \cap K_w^0$. The $K_{bw}^{10}, K_{bb}^{10}, K_{ww}^{10}, K_{wb}^{10}, K_{bw}^{00}, K_{bb}^{00}, K_{ww}^{00}, K_{wb}^{00}, K_{bw}^{11}, K_{bb}^{11}, K_{ww}^{11}, K_{wb}^{11}$ are defined with the similarly way.

We also need the following notations. $K_b^{01} = K_{bw}^{01} \cup K_{bb}^{01}$, $K_w^{01} = K_{wb}^{01} \cup K_{ww}^{01}$. The $K_b^{10}, K_w^{10}, K_b^{00}, K_w^{00}, K_b^{11}, K_w^{11}$ are defined with the similarly way. Let $K^{01} = K_b^{01} \cup K_w^{01}$, $K^{10} = K_b^{10} \cup K_w^{10}$, $K^{00} = K_b^{00} \cup K_w^{00}$ and $K^{11} = K_b^{11} \cup K_w^{11}$. Furthermore, let $K_{bw} = K_{bw}^{01} \cup K_{bw}^{10} \cup K_{bw}^{00} \cup K_{bw}^{11}$ and $K_{bb} = K_{bb}^{01} \cup K_{bb}^{10} \cup K_{bb}^{00} \cup K_{bb}^{11}$, $K_{ww} = K_{ww}^{01} \cup K_{ww}^{10} \cup K_{ww}^{00} \cup K_{ww}^{11}$, $K_{wb} = K_{wb}^{01} \cup K_{wb}^{10} \cup K_{wb}^{00} \cup K_{wb}^{11}$.

3 Vertex fault tolerance for multiple spanning paths in hypercube

Let $dim((x, y)) = i$ if (x, y) is an i -th edge. Let $E' \subset E(Q_n)$ and $DIM(E') = \{dim(e) | e \in E'\}$.

Lemma 1 Let $S \subset V(Q_n)$ where the induced subgraph $\langle S \rangle$ of Q_n is connected. Thus, $|DIM(E(\langle S \rangle))| \leq |S| - 1$.

Proof: We will prove this lemma by induction on $|S|$. When $|S| = 1$, $E(\langle S \rangle) = 0 = |DIM(E(\langle S \rangle))|$. This lemma is true for $|S| = 1$. In the following, we will assume that $|S| \geq 2$. Let $v \in S$ and $S' = S - \{v\}$ where $\langle S' \rangle$ is connected. By induction hypothesis, $|DIM(E(\langle S' \rangle))| \leq |S| - 2$. Suppose that only one vertex in S' is adjacent to v . Thus, $|DIM(E(\langle S \rangle))| \leq |S| - 1$. Suppose that at

least two vertices in S' are adjacent to v . Thus, $|DIM(E(\langle S \rangle))| = |DIM(E(\langle S' \rangle))| \leq |S| - 2$. $\square$

Lemma 2 *Let S be a vertex subset of Q_n where $|S \cap V_w| \geq 1$ and $|S \cap V_b| \geq 1$ for $3 \leq |S| \leq n$. There exist two subcubes Q_{n-1}^0 and Q_{n-1}^1 , such that $d(u, v) \geq 2$ for every pair of vertices $u \in (S \cap V_{n-1}^0)$ and $v \in (S \cap V_{n-1}^1)$. Furthermore, $(|S \cap V_w^i| \geq 1$ and $|S \cap V_b^i| \geq 1)$ for some $i = 0, 1$.*

Proof: Without loss of generality, we can assume that $|S \cap V_w| \geq 2$. Applying Lemma 1, we can obtain that there exists one dimension not in $DIM(E(\langle S \rangle))$. Using this dimension to divide Q_n , we can obtain that $d(u, v) \geq 2$ for every pair of vertices $u \in (S \cap V_{n-1}^0)$ and $v \in (S \cap V_{n-1}^1)$. Assume that $|S \cap V_w^i| \geq 1$ and $|S \cap V_b^i| \geq 1$ is not true. Thus, $|S \cap V_w^i| = 0$ and $|S \cap V_b^i| = 0$ for $0 \leq i \neq j \leq 1$. Hence, $E(\langle S \rangle) = \emptyset$. There exist Q_{n-1}^0 and Q_{n-1}^1 such that $|S \cap V_w^0| \geq 1$ and $|S \cap V_w^1| \geq 1$. Thus, $|S \cap V_w^i| \geq 1$ and $|S \cap V_b^i| \geq 1$ for some $i = 0, 1$. $\square$

Lemma 3 *For $n \geq 3$, if every balanced family $\{s_i, t_i\}_{F_b, K_b}^{F_b, K_b}$ of Q_{2n-2} is connectable where $|F_b| + |F_w| + |K_b| + |K_w| \leq (2n-2)$, $4|F_b| + 2|K_b| \leq (2n-1)$ and $4|F_w| + 2|K_w| \leq (2n-1)$, then every balanced family $\{s_i, t_i\}_{F_w, K_w}^{F_b, K_b}$ of Q_{2n-1} is connectable where $|F_b| + |F_w| + |K_b| + |K_w| \leq (2n-1)$, $4|F_b| + 2|K_b| \leq 2n$ and $4|F_w| + 2|K_w| \leq 2n$.*

Proof: If $|F_b| = |F_w| = 0$, then $|K_b| = |K_w| \leq n-1$ since $\{s_i, t_i\}_{F_w, K_w}^{F_b, K_b}$ is balanced family of Q_{2n-1} . Thus $4|F_b| + 2|K_b| = 4|F_w| + 2|K_w| \leq 2n-1$. The proof for $4|F_b| + 2|K_b| \leq 2n-1$ is the same as in Q_{2n-2} . We will prove this case in Lemma 4. In the following, we will assume that $4|F_b| + 2|K_b| = 4|F_w| + 2|K_w| = 2n$. Thus $|F_b| + |F_w| \geq 1$. Without loss of generality, we can assume that $|F_b| \geq |F_w|$. Thus, $|K_w| \geq |K_b|$. Some cases of Q_{17} are illustrated in the following figures.

Case 1: $|F_w| = 0$ and $(|F^0| = 0$ or $|F^1| = 0)$.

Without loss of generality, we can assume that $|F^1| = 0$.

Case 1.1: $|K_b^1| = 0$.

Case 1.1.1: $|K_w^1| = 0$.

Let $x \in F_b^0$ and $t_1 \in K_w^0$. Since $|F_b^0| - 1 + |K_b^0| + 1 + |K_w^0| - 1 \leq 2n-2$, $4(|F_b^0| - 1) + 2(|K_b^0| + 1) \leq 2n-1$, $4(|F_w^0|) + 2(|K_w^0| - 1) \leq 2n-1$, $\{s_i, t_i\}_{F_b - \{x\}, K_b^0 \cup \{x\}}$ is a

connectable family of Q_{2n-2}^0 . There exist spanning paths $P_1, P_2, \dots, P_{\lfloor \frac{|K|}{2} \rfloor}$ of $Q_{2n-2}^0 - (F_b - \{x\})$. Without loss of generality, we can assume that $P_1 = \langle s_1, \dots, y, x \rangle$ and $P_2 = \langle s_2, \dots, w, t_1, z, \dots, t_2 \rangle$. There exist two spanning paths $\langle \phi(w), \dots, \phi(z) \rangle$ and $\langle \phi(y), \dots, \phi(t_1) \rangle$ in Q_{2n-2}^1 . Thus, there exist $\frac{|K|}{2}$ spanning paths of Q_{2n-1} , as illustrated in Figure 1.(a).

Case 1.1.2: $|K_w^1| \geq 1$.

Applying Lemma 2, we can obtain that $|K_w^0| \geq 1$. Let $x \in F_b^0$ and $U_w^0 \subset (V_w^0 - K_w^0)$ with $|U_w^0| = |K_w^1| - 1$. Since $|F_b^0| - 1 + |K_b^0| + |K_w^1| - 1 + |K_b^0| + 1 \leq 2n-2$, $4(|F_b^0| - 1) + 2(|K_b^0| + 1) \leq 2n-1$, $4|F_w^0| + 2(|K_w^0| - 1) \leq 2n-1$, $\{s_i, t_i\}_{F_b - \{x\}, K_b^0 \cup \{x\}}$ is connectable family of Q_{2n-2}^0 . There exist $\frac{|K|}{2}$ spanning paths between $K_b^0 \cup \{x\} \cup K_w^0 \cup U_w^0$ of Q_{2n-2}^0 . One of these $\frac{|K|}{2}$ paths is $\langle s, \dots, y, x \rangle$. Since $2|K_w^1| \leq 2n-1$, $\{s_i, t_i\}_{\emptyset, K_w^1 \cup \{\phi(y)\}}$ is a connectable family of Q_{2n-2}^1 . There exist $|K_w^1|$ spanning disjoint paths between $\phi(U_w^0) \cup \{\phi(y)\} \cup K_w^1$ of Q_{2n-2}^1 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths of Q_{2n-1} , as illustrated in Figure 1.(b).

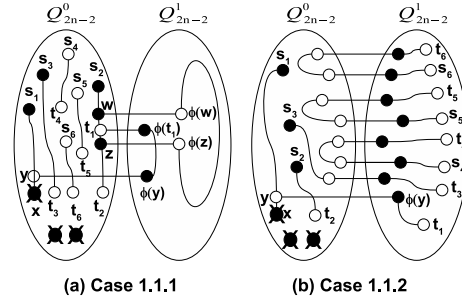


Figure 1: Illustration of Lemma 3

Case 1.2: $|K_b^1| \geq 1, |K_w^1| = 0$.

We use s_1 to denote a white vertex of K_w^{01} if $|K_w^{01}| \geq 1$. When $|K_w^{01}| = 0$, we use s_1 to denote a white vertex of K_w^0 . Let $U_b^0 \subset (V_b^0 - F_b^0 - K_b^0)$ with $|U_b^0| = |K_b^1| - 1$. Since $|F_b^0| + |K_b^0| + |U_b^0| + |K_w^0| - 1 \leq 2n-2$, $4|F_b^0| + 2(|K_b^1| + |U_b^0|) \leq 2n-1$, $2(|K_w^0| - 1) \leq 2n-1$, $\{s_i, t_i\}_{F_b - \{s_1\}, K_b^0 \cup U_b^0}$ is a connectable family of Q_{2n-2}^0 . There exist $\frac{|K|}{2} - 1$ spanning paths of Q_{2n-2}^0 . Without loss of generality, we can assume that one of

these paths is $\langle s_2, \dots, y, s_1, z, \dots, t_2 \rangle$. Since $2(|K_b^1| + 1) \leq 2n - 1$, $\{s_i, t_i\}_{\emptyset, K_b^1 \cup \{\phi(s_1)\}}^{\emptyset, K_b^1 \cup \{\phi(s_1)\}}$ is a connectable family of Q_{2n-2}^1 . There exist $|K_b^1| + 1$ spanning paths in Q_{2n-2}^1 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} , as illustrated in Figure 2.(a).

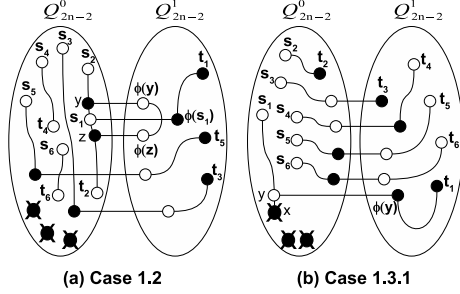


Figure 2: Illustration of Lemma 3

Case 1.3: $|K_b^1| \geq 1, |K_w^1| \geq 1$.

Case 1.3.1: $|K^{11}| = 0, |K_w^1| \geq |K_b^1|$.

Let $s_1 \in K^0, t_1 \in K_b^1$ and $x \in F_b^0$. Let $U_w^{01} \subset (V_w^0 - K_w^0)$ for $\phi(U_w^{01}) = (K_b^1 - \{t_1\})$. Let $U_w^0 \subset (V_w^0 - K_w^0 - U_w^{01})$ with $\phi(U_w^0) \subset (V_b^1 - K_b^1)$, $|U_w^0| = (n - 1 - |K_w^0| - |U_w^{01}|)$ and $U_b^0 \subset (V_b^0 - F_b^0 - K_b^0)$ with $\phi(U_b^0) \subset (V_w^1 - K_w^1)$, $|U_b^0| = (n - 2|F_b^0| - |K_b^0|)$. Since $|F_b^0| + |K_b^0| + |U_b^0| + |K_w^0| + |U_w^{01}| + |U_w^0| = 2n - 1 - |F_b^0| < 2n - 1$, $4(|F_b^0| - 1) + 2(|K_b^0| + 1 + |U_b^0|) = 2n - 2$, $2(|K_w^0| + |U_w^{01}| + |U_w^0|) = 2n - 2$, $\{s_i, t_i\}_{\emptyset, K_b^0 \cup \{x\} \cup U_b^0}^{F_b^0 - \{x\}, K_b^0 \cup \{x\} \cup U_b^0}$ is a connectable family of Q_{2n-2}^0 . There exist $\frac{|K|}{2}$ spanning paths of Q_{2n-2}^0 . One of these paths is $\langle s_1, \dots, y, x \rangle$.

Suppose $\phi(y) \neq t_1$. Since $|K_b^1| + |K_w^1| + |\phi(U_w^0)| + |\phi(U_b^0)| \leq 2n - 1 - |U_w^{01}| - 2|F_b^0| < 2n - 1$, $4(|K_b^1| - 1) + 2(2 + 2|\phi(U_w^0)|) = 2n + 2|K_b^1| - 2|K_w^0| \leq 2n - 2$ and $2|K_w^1| + 2|\phi(U_b^0)| = 2n - 4|F_b^0| - 2|K_b^0| + 2|K_b^1| = 2n + 2|K_b^1| - 2|K_w^0| \leq 2n - 2$, $\{s_i, t_i\}_{\emptyset, K_b^1 - \{t_1\}, \{\phi(y), t_1\} \cup \phi(U_w^0)}^{K_b^1 - \{t_1\}, \{\phi(y), t_1\} \cup \phi(U_w^0)}$ is a connectable family of Q_{2n-2}^1 . There exist $(n - |K_w^0| + 1)$ spanning paths of Q_{2n-2}^1 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} , as illustrated in Figure 3.(a).

Suppose $\phi(y) = t_1$. Since $|K_b^1| + |K_w^1| + |\phi(U_w^0)| + |\phi(U_b^0)| \leq 2n - 1 - |U_w^{01}| - 2|F_b^0| < 2n - 1$, $4|K_b^1| + 2(2|\phi(U_w^0)|) = 2n + 2|K_b^1| - 2|K_w^0| \leq 2n - 2$ and $2|K_w^1| + 2|\phi(U_b^0)| = 2n - 4|F_b^0| - 2|K_b^0| + 2|K_b^1| = 2n + 2|K_b^1| - 2|K_w^0| \leq$

$2n - 2$, $\{s_i, t_i\}_{\emptyset, K_b^1 \cup \phi(U_w^0)}^{K_b^1 \cup \phi(U_w^0)}$ is a connectable family of Q_{2n-2}^1 . There exist $(n - |K_w^0|)$ spanning paths of Q_{2n-2}^1 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} .

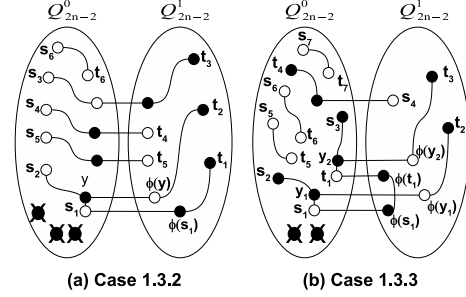


Figure 3: Illustration of Lemma 3

Case 1.3.2: $|K^{11}| = 0, |K_b^1| > |K_w^1|$ and $|K_w^{01}| \geq 1$

Let $s_1 \in K_w^{01}$ and $s_2 \in K^{01}$. Let $U_b^{01} \subset (V_b^0 - F_b^0 - K_b^0)$ for $\phi(U_b^{01}) = K_w^1$. And let $U_w^0 \subset (V_w^0 - K_w^0)$ with $\phi(U_w^0) \subset (V_b^1 - K_b^1)$, $|U_w^0| = (|K_w^1| - 1)$ and $U_b^0 \subset (V_b^0 - F_b^0 - K_b^0 - U_b^{01})$ with $|U_b^0| = (n - 1 - 2|F_b^0| - |K_b^0| - |U_b^{01}|)$. Since $|F_b^0| + |K_b^0| + |U_b^{01}| + |U_b^0| + |K_w^0| + |U_w^0| = 2n - 2 - |F_b^0| < 2n - 1$, $4|F_b^0| + 2(|K_b^0| + |U_b^{01}| + |U_b^0|) = 2n - 2 = 2(|K_w^0| + |U_w^0|)$, $\{s_i, t_i\}_{\emptyset, K_b^0 \cup U_b^{01} \cup U_b^0}^{F_b^0 - K_b^0 \cup U_b^{01} \cup U_b^0}$ is a connectable family of Q_{2n-2}^0 . There exist $\frac{|K|}{2} - 1$ spanning paths of Q_{2n-2}^0 . One of these paths is $\langle s_2, \dots, y, s_1 \rangle$. Since $|K_b^1| + |\phi(U_w^0)| + |K_w^1| + |\phi(U_b^0)| + 2 = n + 2|K_b^1| - |K_w^0| < n + |K_b^1| \leq 2n - 1$, $2|K_b^1| + 2 + 2(|\phi(U_w^0)|) = 2|K_b^1| + 2|K_w^1| \leq 2n - 2$ and $4|K_w^1| + 2 + 2|\phi(U_b^0)| = 2|K_b^1| + 2|K_w^1| \leq 2n - 2$, $\{s_i, t_i\}_{\emptyset, K_b^1 \cup \{\phi(s_2)\} \cup \phi(U_w^0)}^{K_b^1 \cup \{\phi(s_2)\} \cup \phi(U_w^0)}$ is a connectable family of Q_{2n-2}^1 . There exist $|K_b^1|$ spanning paths of Q_{2n-2}^1 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} , as illustrated in Figure 3.(b).

Case 1.3.3: $|K^{11}| = 0, |K_b^1| > |K_w^1|$ and $|K_w^{01}| = 0$

Since $|K^{11}| = 0 = |K_w^{01}|$, $|K_w^{00}| = |K_w^0| - |K_w^{00}| \geq 2|F_b^0| + 2|K_b^1| \geq 2$. Let $s_1, t_1 \in K_w^{00}$ and $s_2, s_3 \in K_b^{01}$. Let $U_b^{01} \subset (V_b^0 - F_b^0 - K_b^0)$ for $\phi(U_b^{01}) = K_w^1$. And let $U_w^0 \subset (V_w^0 - K_w^0)$ with $\phi(U_w^0) \subset (V_b^1 - K_b^1)$, $|U_w^0| = (|K_w^1| - 1)$ and $U_b^0 \subset (V_b^0 - F_b^0 - K_b^0 - U_b^{01})$ with $\phi(U_b^0) \subset (V_w^1 - K_w^1)$, $|U_b^0| = (n - 1 - 2|F_b^0| - |K_b^0| - |U_b^{01}|)$. Since $|F_b^0| + |K_b^0| + |U_b^{01}| +$

$|U_b^0| + |K_w^0| + |U_w^0| = 2n - 2 - |F_b^0| < 2n - 1$, $4|F_b^0| + 2(|K_b^0| + |U_b^0| + |U_w^0|) = 2n - 2 = 2(|K_w^0| + |U_w^0|)$, $\{s_i, t_i\}_{\emptyset, K_b^0 \cup U_b^0 \cup U_w^0}$ is a connectable family of Q_{2n-2}^0 . There exist $\frac{|K|}{2} - 1$ spanning paths of Q_{2n-2}^0 . Two of these paths are $\langle s_2, \dots, y_1, s_1 \rangle$ and $\langle s_3, \dots, y_2, t_1 \rangle$. Since $|K_b^1| + |\phi(U_w^0)| + |K_w^1| + |\phi(U_b^0)| + 4 = |K_b^1| + |K_w^1| - 1 + |K_w^1| + (n - 1 - 2|F_b^0| - |K_b^0| - |K_w^1|) = n - 2|F_b^0| + 2 + |K_b^1| + |K_w^1| - |K_b^0| \leq n - 2|F_b^0| + 2 < 2n - 1$, $2|K_b^1| + 4 + 2(|\phi(U_w^0)| + |\phi(U_b^0)|) = 2|K_b^1| + 2|K_w^1| + 2 \leq 2|K_b^0| + 2 \leq 2n - 2$ and $4|K_w^1| + 4 + 2|\phi(U_b^0)| = 2|K_b^1| + 2|K_w^1| + 2 \leq 2|K_b^0| + 2 \leq 2n - 2$, $\{s_i, t_i\}_{K_b^1 \cup \{\phi(s_1), \phi(t_1)\} \cup \phi(U_w^0)}$ is a connectable family of Q_{2n-2}^1 . There exist $|K_b^1| + 1$ spanning paths of Q_{2n-2}^1 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} , as illustrated in Figure 3.(b).

Case 1.3.4: $|K^{11}| \geq 2, |K_w^0| \geq 1$.

Let $U_w^0 \subset (V_w^0 - K_w^0)$ with $\phi(U_w^0) \subset (V_b^1 - K_b^1)$, $|U_w^0| = \min(|K_w^1| - 1, |K_b^0| + 2|F_b^0| - 1)$ and $U_b^0 \subset (V_b^0 - F_b^0 - K_b^0)$ with $\phi(U_b^0) \subset (V_w^1 - K_w^1)$, $|U_b^0| = \min(|K_b^1| - 1, |K_w^0| - 1)$. Since $|K_w^1| = n - |K_w^0| = |K_b^0| + 2|F_b^0| + |K_b^1| - |K_w^0|$, $|K_b^1| - 1 \leq |K_w^0| - 1$ if and only if $|K_w^1| - 1 \leq |K_b^0| + 2|F_b^0|$. Since $|F_b^0| + |K_b^0| + |U_b^0| + |K_w^0| + |U_w^0| = 2n - 2 - |F_b^0| < 2n - 1$, $4|F_b^0| + 2(|K_b^0| + |U_b^0|) = 2n - 2 = 2(|K_w^0| + |U_w^0|) = 2(|K_w^0| + |U_w^0|) \leq 2n - 2$, $\{s_i, t_i\}_{\emptyset, K_b^0 \cup U_b^0 \cup U_w^0}$ is a connectable family of Q_{2n-2}^0 . There exist $\frac{|K_b^0| + |U_b^0| + |K_w^0| + |U_w^0|}{2}$ spanning paths of Q_{2n-2}^0 . Since $|K_b^1| + |\phi(U_w^0)| + |K_w^1| + |\phi(U_b^0)| \leq 2n - 2$, $2|K_b^1| + 2|\phi(U_w^0)| = 2|K_w^1| + 2|\phi(U_b^0)| \leq 2n - 2$, $\{s_i, t_i\}_{\emptyset, K_b^1 \cup \phi(U_w^0)}$ is a connectable family of Q_{2n-2}^1 . There exist $(|K_b^1| + |K_w^1| + |U_b^0| + |U_w^0|)/2$ spanning paths of Q_{2n-2}^1 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} , as illustrated in Figure 4.(a).

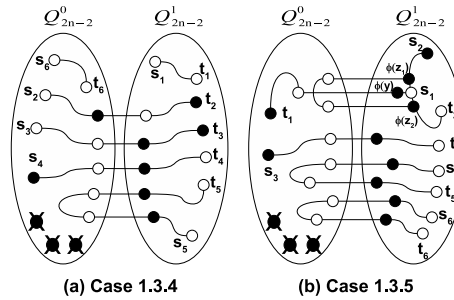


Figure 4: Illustration of Lemma 3

Case 1.3.5: $|K^{11}| \geq 2, |K_w^0| = 0$.

Let $s_1 \in K_w^1$ and $\phi(y) \in (V_b^1 - K_b^1)$ be a neighbor of s_1 . Let $\phi(U_w^0) \subset (V_b^1 - K_b^1)$ with $|U_w^0| = n - 3 - |K_b^1|$. Since $|K_b^1| + |\phi(U_w^0)| + |K_w^1| \leq 2n - 3$, $4|K_b^1| + 2|\phi(U_w^0)| = 2n - 2 = 2(|K_w^1| - 1)$, $\{s_i, t_i\}_{\emptyset, K_w^1 - \{s_1\}}$ is a connectable family of Q_{2n-2}^1 . There exist $(n - 2)$ spanning paths of Q_{2n-2}^1 . Without loss of generality, we can assume that one of these paths is $\langle s_2, \dots, \phi(z_1), s_1, \phi(z_2), \dots, t_2 \rangle$. Since $|F_b^0| + |K_b^0| + |U_b^0| + 3 \leq 2n - 2$, $4|F_b^0| + 2(|K_b^0| + |U_b^0|) = 2n - 2 = 2(|U_b^0| + 3) \leq 2n - 2$, $\{s_i, t_i\}_{\emptyset, \{y, z_1, z_2\} \cup U_b^0}$ is a connectable family of Q_{2n-2}^0 . There exist $(n - |F_b^0|)$ spanning paths of Q_{2n-2}^0 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} , as illustrated in Figure 4.(b).

Case 2: $|F_w| = 0, |F_b^0| \geq 1, |F_b^1| \geq 1$.

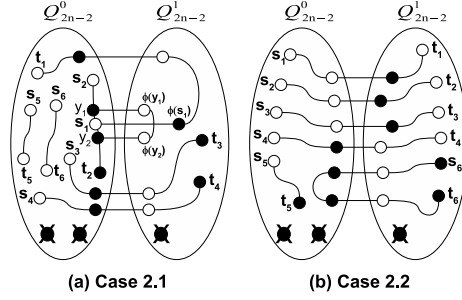
Case 2.1: $K_w \subset V_w^0$ or $K_w \subset V_w^1$.

Without loss of generality, we can assume that $K_w \subset V_w^0$. Let $s_1 \in K_w^0$ and $U_b^0 \subset (V_b^0 - F_b^0 - K_b^0)$ with $|U_b^0| = n - 1 - 2|F_b^0| - |K_b^0|$. Since $|F_b^0| + |K_b^0| + |U_b^0| + |K_w^0| - 1 = n - 1 + |K_w^0| - |F_b^0| = 2n - 1 - |F_b^0| < 2n - 1$, $4|F_b^0| + 2(|K_b^0| + |U_b^0|) = 2n - 2 = 2(|K_w^0| - 1)$, $\{s_i, t_i\}_{\emptyset, K_b^0 \cup U_b^0 - \{s_1\}}$ is a connectable family of Q_{2n-2}^0 . There exist $\frac{|K|}{2}$ spanning paths of Q_{2n-2}^0 . Without loss of generality, we can assume that one of these paths is $\langle s_2, \dots, y_1, s_1, y_2, \dots, t_2 \rangle$. Since $|F_b^1| + |K_b^1| + |\phi(U_b^0)| + 3 = n + 2 + |F_b^1| + |K_b^1| - 2|F_b^0| - |K_b^0| = 2n + 2 - |F_b^1| - 4|F_b^0| - 2|K_b^0| \leq 2n - 3$, $4|F_b^1| + 2|K_b^1| + 2 = 2|\phi(U_b^0)| + 4 \leq 2n - 2$, $\{s_i, t_i\}_{\emptyset, \phi(U_b^0) \cup \{\phi(y_1), \phi(y_2)\}}$ is a connectable family of Q_{2n-2}^1 . There exist $(|K_b^1| + 1 + |F_b^1|)$ spanning paths of Q_{2n-2}^1 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} , as illustrated in Figure 5.(a).

Case 2.2: $|K_w^0| \geq 1, |K_w^1| \geq 1$.

Let $U_w^0 \subset (V_w^0 - K_w^0)$ with $\phi(U_w^0) \subset (V_b^1 - F_b^1 - K_b^1)$, $|U_w^0| = \min(|K_w^1| - 1, |K_b^0| + 2|F_b^0| - 1)$ and $U_b^0 \subset (V_b^0 - F_b^0 - K_b^0)$ with $\phi(U_b^0) \subset (V_w^1 - K_w^1)$, $|U_b^0| = \min(|K_b^1| + 2|F_b^1| - 1, |K_w^0| - 1)$. Since $|K_w^1| + |K_w^0| = n = |K_b^0| + 2|F_b^0| + |K_b^1| + 2|F_b^1|$, $|K_w^1| - 1 \leq |K_b^0| + 2|F_b^0| - 1$ if and only if $|K_b^1| + 2|F_b^1| - 1 \leq |K_w^0| - 1$. Since $|F_b^0| + |K_b^0| + |U_b^0| + |K_w^0| + |U_w^0| \leq 2n - 2 - |F_b^0| \leq 2n - 3$, $4|F_b^0| + 2(|K_b^0| + |U_b^0|) = 2(|K_w^0| + |U_w^0|) \leq 2n - 2$, $\{s_i, t_i\}_{\emptyset, K_b^0 \cup U_b^0 \cup U_w^0}$

is a connectable family of Q_{2n-2}^0 . There exist $\frac{|K_b^0|+|U_b^0|+|K_w^0|+|U_w^0|}{2}$ spanning paths of Q_{2n-2}^0 . Since $|F_b^1|+|K_b^1|+|\phi(U_w^0)|+|K_w^1|+|\phi(U_b^0)| \leq 2n-2-|F_b^1| \leq 2n-3, 4|F_b^1|+2(|K_b^1|+|\phi(U_b^0)|) = 2(|K_w^1|+|\phi(U_b^0)|) \leq 2n-2, \{s_i, t_i\}_{\emptyset, K_w^1 \cup \phi(U_b^0)}$ is a connectable family of Q_{2n-2}^1 . There exist $\frac{|K_b^1|+|U_b^1|+|K_w^1|+|U_w^1|}{2}$ spanning paths of Q_{2n-2}^1 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} , as illustrated in Figure 5.(a).


 Figure 5: Illustration of **Lemma 3**

Case 3: $|F_b| \geq 1, |F_w| \geq 1$ and $(|F^0| = 0$ or $|F^1| = 0)$.

Without loss of generality, we can assume that $|F^1| = 0$.

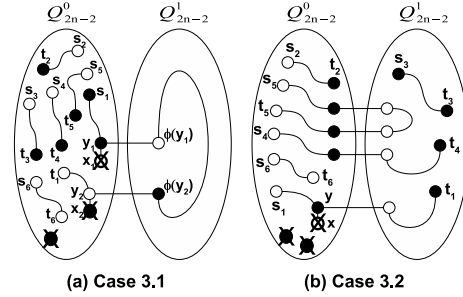
Case 3.1: $|K^1| = 0$.

Let $x_1 \in F_b^0$ and $x_2 \in F_w^0$. Since $|F_b^0|+|K_b^0|+|F_w^0|+|K_w^0| = 2n-|F_b^0|-|F_w^0| \leq 2n-2, 4(|F_b^0|-1)+2(|K_b^0|+1) = 2n-2 = 4(|F_w^0|-1)+2(|K_w^0|+1), \{s_i, t_i\}_{F_b^0-\{x_1\}, K_b^0 \cup \{x_1\}}$ is a connectable family of Q_{2n-2}^0 . There exist $\frac{|K|}{2}+1$ spanning paths of Q_{2n-2}^0 . Without loss of generality, we can assume that two of these paths are $\langle s_1, \dots, y_1, x_1 \rangle$ and $\langle t_1, \dots, y_2, x_2 \rangle$. There is a Hamiltonian path $\langle \phi(y_1), \dots, \phi(y_2) \rangle$. Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} , as illustrated in Figure 6.(a).

Case 3.2: $|K_w^1| = 0$ or $|K_b^1| = 0$.

Without loss of generality, we can assume that $|K_w^1| = 0$. Let $x \in F_w^0$. We use s_1 to denote a white vertex of K_w^{01} if $|K_w^{01}| \geq 1$. When $|K_w^{01}| = 0$, we use s_1 to denote a vertex of K^0 . Let $U_b^0 \subset (V_b^0 - F_b^0 - K_b^0)$ with $|U_b^0| = |K_b^1| - 1$. Since $|F_b^0|+|K_b^0|+|U_b^0|+|F_w^0|+|K_w^0| = 2n-1-|F_w^0|-|F_b^0| \leq$

$2n-3, 4|F_b^0|+2(|K_b^0|+|U_b^0|) = 2n-2 = 4(|F_w^0|-1)+2(K_w^0+1), \{s_i, t_i\}_{F_b^0-\{x\}, K_b^0 \cup U_b^0}$ is a connectable family of Q_{2n-2}^0 . There exist $\frac{|K|}{2}$ spanning paths of Q_{2n-2}^0 . One of these paths is $\langle s_1, \dots, y, x \rangle$. Since $|K_b^1|+|\phi(U_w^0)|+1 \leq 2n-2, 2|K_b^1| = 2(|\phi(U_b^0)|+1) \leq 2n-2, \{s_i, t_i\}_{\emptyset, K_b^1 \cup \phi(U_b^0)}$ is a connectable family of Q_{2n-2}^1 . There exist $\frac{|K_b^1|}{2}$ spanning paths of Q_{2n-2}^1 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} , as illustrated in Figure 6.(b).


 Figure 6: Illustration of **Lemma 3**

Case 3.3: $|K_w^1| \geq 1, |K_b^1| \geq 1$.

Case 3.3.1: $|K^{11}| = 0$

Without loss of generality, we assume that $|K_b^1| \geq |K_w^1|$. Let $t_1 \in K_w^1, x \in F_w^0$ and $U_b^{01} \subset (V_b^0 - F_b^0 - K_b^0)$ with $\phi(U_b^{01}) = (K_w^1 - \{t_1\})$. Let $U_w^0 \subset (V_w^0 - F_w^0 - K_w^0)$ with $\phi(U_w^0) \subset (V_b^1 - K_b^1), |U_w^0| = (n-|K_w^0|-2|F_w^0|)$ and $U_b^0 \subset (V_b^0 - F_b^0 - K_b^0 - U_b^{01})$ with $\phi(U_b^0) \subset (V_w^1 - K_w^1), |U_b^0| = (n-2|F_b^0|-|K_b^0|-|K_w^1|)$. Since $|F_b^0|+|K_b^0|+|U_b^{01}|+|U_b^0|+|F_w^0|+|K_w^0|+|U_w^0| = 2n-1-|F_w^0|-|F_b^0| \leq 2n-3, 4|F_b^0|+2(|K_b^0|+|U_b^{01}|+|U_b^0|) = 2n-2 = 4(|F_w^0|-1)+2(K_w^0+1+|U_w^0|), \{s_i, t_i\}_{F_b^0-\{x\}, K_b^0 \cup U_b^{01} \cup U_b^0}$ is a connectable family of Q_{2n-2}^0 . There exist $\frac{|K|}{2}$ spanning paths of Q_{2n-2}^0 . One of these paths is $\langle s_1, \dots, y, x \rangle$. Since $|K_b^1|+|\phi(U_w^0)|+|K_w^1|+|\phi(U_b^0)|+1 = 2|K_b^1|+|K_w^1|+1 \leq 2n-4, 2(|K_b^1|+|\phi(U_w^0)|) = 2(|K_b^1|+|K_w^1|) = 4(|K_w^1|-1)+4+2|\phi(U_b^0)| \leq 2n-4, \{s_i, t_i\}_{K_b^1 \cup \phi(U_w^0), \phi(U_b^0) \cup \phi(y), t_1}$ is a connectable family of Q_{2n-2}^1 . There exist $(|K_b^1|+1)$ spanning paths of Q_{2n-2}^1 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} , as illustrated in Figure 7.(a).

Case 3.3.2: $K^{11} \geq 2$

Let $U_w^0 \subset (V_w^0 - F_w^0 - K_w^0)$ with $\phi(U_w^0) \subset (V_b^1 - K_b^1)$, $|U_w^0| = \min(|K_w^1| - 1, |K_b^0| + 2|F_b^0| - 1)$ and $U_b^0 \subset (V_b^0 - F_b^0 - K_b^0)$ with $\phi(U_b^0) \subset (V_w^1 - K_w^1)$, $|U_b^0| = \min(|K_b^1| - 1, |K_w^0| + 2|F_w^0| - 1)$. Since $|K_w^1| + 2|F_w^0| + |K_b^0| = n = |K_b^0| + 2|F_b^0| + |K_w^1| + 2|F_b^1|$, $|K_w^1| - 1 \leq |K_b^0| + 2|F_b^0| - 1$ if and only if $|K_b^1| - 1 \leq |K_w^0| + 2|F_w^0| - 1$. Since $|F_b^0| + |K_b^0| + |U_b^0| + |F_w^0| + |K_w^0| + |U_w^0| \leq 2n - 2 - |F_b^0| - |F_w^0| \leq 2n - 4$, $4|F_b^0| + 2(|K_b^0| + |U_b^0|) = 4|F_w^0| + 2(|K_w^0| + |U_w^0|) \leq 2n - 2$, $\{s_i, t_i\}_{F_b^0, K_b^0 \cup U_b^0 \cup U_w^0}$ is a connectable family of Q_{2n-2}^0 . There exist $\frac{|K_b^0| + |U_b^0| + |K_w^0| + |U_w^0|}{2}$ spanning paths of Q_{2n-2}^0 . Since $|K_b^1| + |\phi(U_w^0)| + |K_w^1| + |\phi(U_b^0)| \leq 2n - 2$, $2(|K_b^1| + |\phi(U_w^0)|) = 2(|K_w^1| + |\phi(U_b^0)|) \leq 2n - 2$, $\{s_i, t_i\}_{\emptyset, K_b^1 \cup \phi(U_w^0)}$ is a connectable family of Q_{2n-2}^1 . There exist $\frac{|K_b^1| + |U_b^0| + |K_w^1| + |U_w^0|}{2}$ spanning paths of Q_{2n-2}^1 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} , as illustrated in Figure 7.(b).

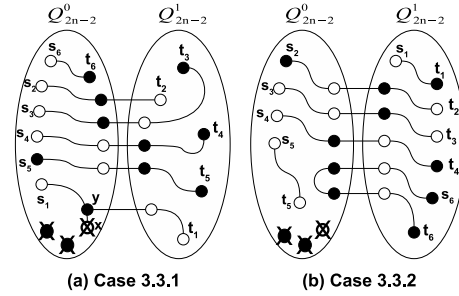


Figure 7: Illustration of Lemma 3

Case 4: $|F_b| \geq 1, |F_w| \geq 1, |F^0| \geq 1, |F^1| \geq 1$.

Case 4.1: $|K_b^0| + |F_b^0| = 0$ or $|K_w^0| + |F_w^0| = 0$ or $|K_b^1| + |F_b^1| = 0$ or $|K_w^1| + |F_w^1| = 0$.

Without loss of generality, we assume that $|K_b^1| + |F_b^1| = 0$. We use s_1 to denote a vertex of K^{01} if $|K^{01}| \geq 1$. When $|K^{01}| = 0$, we use s_1 to denote a vertex of K^{00} . Let $x \in F_w^0$. Let $U_b^0 \subset (V_b^0 - F_b^0 - K_b^0)$ with $|U_b^0| = (2|F_b^1| + |K_b^1| - 1)$. Since $|F_b^0| + |K_b^0| + |U_b^0| + |F_w^0| + |K_w^0| = 2n - 1 - |F_b^0| - |F_w^0| \leq 2n - 2$, $4|F_b^0| + 2(|K_b^0| + |U_b^0|) = 2n - 2 = 4(|F_w^0| - 1) + 2(|K_w^0| + 1)$, $\{s_i, t_i\}_{F_b^0, K_b^0 \cup U_b^0 \cup \{x\}}$ is a connectable family of Q_{2n-2}^0 . There exist $\frac{|K_b^1| + |K_b^0| + |K_w^0| + 2|F_b^1|}{2}$ spanning paths of Q_{2n-2}^0 . One of these paths is $\langle s_1, \dots, y, x \rangle$. Since $|F_b^1| + |K_b^1| + |\phi(U_b^0)| + 1 = 2n - |F_b^1| -$

$2|K_b^0| - 4|F_b^0| \leq 2n - 3$, $4|F_b^1| + 2(|K_b^1| + 2|\phi(U_b^0)| + 2 \leq 2n - 2$, $\{s_i, t_i\}_{F_b^1, K_b^1 \cup \phi(U_b^0) \cup \{y\}}$ is a connectable family of Q_{2n-2}^1 . There exist $(|K_b^1| + |F_b^1|)$ spanning paths of Q_{2n-2}^1 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} , as illustrated in Figure 8.(a).

Case 4.2: $|K_b^0| + |F_b^0| \geq 1$ and $|K_w^0| + |F_w^0| \geq 1$ and $|K_b^1| + |F_b^1| \geq 1$ and $|K_w^1| + |F_w^1| \geq 1$.

Let $U_w^0 \subset (V_w^0 - F_w^0 - K_w^0)$ with $\phi(U_w^0) \subset (V_b^1 - K_b^1)$, $|U_w^0| = \min(|K_w^1| + 2|F_w^1| - 1, |K_b^0| + 2|F_b^0| - 1)$ and $U_b^0 \subset (V_b^0 - F_b^0 - K_b^0)$ with $\phi(U_b^0) \subset (V_w^1 - F_w^1 - K_w^1)$, $|U_b^0| = \min(|K_b^1| + 2|F_b^1| - 1, |K_w^0| + 2|F_w^0| - 1)$. Since $|K_w^0| + 2|F_w^0| + |K_w^1| + 2|F_w^1| = n = |K_b^0| + 2|F_b^0| + |K_b^1| + 2|F_b^1|$, $|K_w^1| + 2|F_w^1| - 1 \leq |K_b^0| + 2|F_b^0| - 1$ if and only if $|K_b^1| + 2|F_b^1| - 1 \leq |K_w^0| + 2|F_w^0| - 1$. Since $|F_b^0| + |K_b^0| + |U_b^0| + |F_w^0| + |K_w^0| + |U_w^0| \leq 2n - 2 - |F_b^0| - |F_w^0| \leq 2n - 2$, $4|F_b^0| + 2(|K_b^0| + |U_b^0|) = 4|F_w^0| + 2(|K_w^0| + |U_w^0|) \leq 2n - 2$, $\{s_i, t_i\}_{F_b^0, K_b^0 \cup U_b^0 \cup U_w^0}$ is a connectable family of Q_{2n-2}^0 . There exist $\frac{|K_b^0| + |U_b^0| + |K_w^0| + |U_w^0|}{2}$ spanning paths of Q_{2n-2}^0 . Since $|F_b^1| + |K_b^1| + |\phi(U_w^0)| + |F_w^1| + |K_w^1| + |\phi(U_b^0)| \leq 2n - 2 - |F_b^1| - |F_w^1| \leq 2n - 2$, $4|F_b^1| + 2(|K_b^1| + |\phi(U_w^0)|) = 4|F_w^1| + 2(|K_w^1| + |\phi(U_b^0)|) \leq 2n - 2$, $\{s_i, t_i\}_{F_b^1, K_b^1 \cup \phi(U_w^0) \cup \phi(U_b^0)}$ is a connectable family of Q_{2n-2}^1 . There exist $\frac{|K_b^1| + |U_b^0| + |K_w^1| + |U_w^0|}{2}$ spanning paths of Q_{2n-2}^1 . Therefore, we can construct $\frac{|K|}{2}$ spanning paths in Q_{2n-1} , as illustrated in Figure 8.(b). $\square$

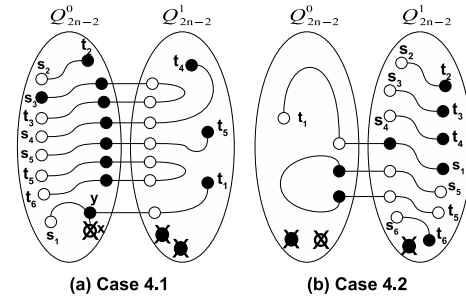


Figure 8: Illustration of Lemma 3

Lemma 4 If every balanced family $\{s_i, t_i\}_{F_b, K_b}$ of Q_{2n-1} is connectable where $|F_b| + |F_w| + |K_b| + |K_w| \leq (2n - 1)$, $4|F_b| + 2|K_b| \leq 2n$ and $4|F_w| + 2|K_w| \leq 2n$, then every balanced family $\{s_i, t_i\}_{F_b, K_b}$ of Q_{2n} is connectable where $|F_b| +$

$|F_w| + |K_b| + |K_w| \leq (2n), 4|F_b| + 2|K_b| \leq 2n + 1$ and $4|F_w| + 2|K_w| \leq 2n + 1$, for $n \geq 2$.

Proof: Since $4|F_b| + 2|K_b| \leq 2n + 1$ and $4|F_b| + 2|K_b|$ is even, $4|F_b| + 2|K_b| \leq 2n$. Similarly, $4|F_w| + 2|K_w| \leq 2n$. Thus, $4|F_b| + 2|K_b| + 4|F_w| + 2|K_w| \leq 4n$. Therefore, $|F_b| + |F_w| + |K_b| + |K_w| \leq 2n - |F_b| - |F_w|$. If $|F_b| + |F_w| \geq 1$ or $|K_w| + |K_b| \leq 2n - 1$, then $|F_b| + |F_w| + |K_b| + |K_w| \leq 2n - 1$. The proof of this case is similar to the proof of Lemma 3. In the following, we will assume that $|F_b| = |F_w| = 0$ and $|K_w| = |K_b| = n$. Without loss of generality, we can assume that $|K^0| \geq |K^1|$. Some cases of Q_{18} are illustrated in the following figures.

Case 1: $|K_b^0| = n$ or $|K_w^0| = n$. Without loss of generality, we can assume that $|K_b^0| = n$.

Case 1.1: $|K_w^0| = n$.

Case 1.1.1: $|K_{bb}^{00}| \geq 1$.

Let $s_1, t_1 \in K_b^0$. Suppose that $n = 2$. There exists a Hamiltonian path $\langle s_2, \dots, y_1, t_1, y_2, y_3, \dots, t_2 \rangle$ of $Q_3^0 - \{s_1\}$. Applying Lemma 1 of [5], we can obtain that there exist two spanning paths $\langle \phi(y_1), \dots, \phi(y_3) \rangle$ and $\langle \phi(s_1), \dots, \phi(y_2) \rangle$ of Q_3^1 . Therefore, $\langle s_1, \phi(s_1), \dots, \phi(y_2), y_2, t_1 \rangle$ and $\langle s_2, \dots, y_1, \phi(y_1), \dots, \phi(y_3), y_3, \dots, t_2 \rangle$ are two spanning paths of Q_4 .

Suppose that $n \geq 3$. Since $|K_b^0| - 1 + |K_w^0| = 2n - 1, 4 + 2(|K_b^0| - 2) = 2n = 2|K_w^0|$, $\{s_i, t_i\}_{\emptyset, K_b^0 - \{s_1, t_1\}}$ is a connectable family of Q_{2n-1}^0 . There exist $\frac{|K|}{2} - 1$ spanning paths of Q_{2n-1}^0 . Without loss of generality, we can assume that one of these paths is $\langle s_2, \dots, y_1, t_1, y_2, \dots, t_2 \rangle$. There exist two spanning paths $\langle \phi(y_1), \dots, \phi(y_2) \rangle$ and $\langle \phi(s_1), \dots, \phi(t_1) \rangle$. Therefore, we can construct n spanning paths in Q_{2n} , as illustrated in Figure 9.(a).

Case 1.1.2: $|K_{bb}^{00}| = 0$.

Let $s_1, s_2 \in K_b^0$. Suppose that $n = 2$. There exists a Hamiltonian path $\langle t_1, \dots, y, s_2, \dots, t_2 \rangle$ of $Q_3^0 - \{s_1\}$. There exists a Hamiltonian path $\langle \phi(s_1), \dots, \phi(y) \rangle$ of Q_3^1 . Therefore, $\langle s_1, \phi(s_1), \dots, \phi(y), y, \dots, t_1 \rangle$ and $\langle s_2, \dots, t_2 \rangle$ are two spanning paths of Q_4 .

Suppose that $n = 3$. If none of K_w^0 is adjacent to some vertex of K_b^0 , we can arrange some vertices of K in both Q_5^0 and Q_5^1 . We can assume that s_1 is adjacent to $t_2 \in K_w^0$. Applying Theorem

2 of [8], we can obtain that there exist two spanning paths $\langle s_2, \dots, y_1, y_2, \dots, t_1 \rangle$ and $\langle s_3, \dots, t_3 \rangle$ in $Q_5^0 - \{s_1, t_2\}$. There exist two spanning paths $\langle \phi(y_1), \dots, \phi(t_2) \rangle$ and $\langle \phi(s_1), \dots, \phi(y_2) \rangle$ in Q_5^1 . Therefore, $\langle s_1, \phi(s_1), \dots, \phi(y_2), y_2, \dots, t_1 \rangle$, $\langle s_2, \dots, y_1, \phi(y_1), \dots, \phi(t_2), t_2 \rangle$ and $\langle s_3, \dots, t_3 \rangle$ are three spanning paths of Q_6 .

Suppose that $n \geq 4$. Since $\{s_i, t_i\}_{\emptyset, K_b^0 - \{s_1\}}$ is a connectable family, there exist $\frac{|K|}{2} - 1$ spanning paths of Q_{2n-1}^0 . Suppose s_1 and t_1 are on different paths. Without loss of generality, we can assume that two of these paths are $\langle s_2, \dots, y_1, s_1, y_2, \dots, t_2 \rangle$ and $\langle s_3, \dots, z_1, t_1, z_2, \dots, t_3 \rangle$. Since $\{s_i, t_i\}_{\emptyset, \{\phi(t_1), \phi(y_1), \phi(y_2)\}}$ is a connectable family, there exist three spanning disjoint paths $\langle \phi(s_1), \dots, \phi(t_1) \rangle$, $\langle \phi(y_1), \dots, \phi(y_2) \rangle$ and $\langle \phi(z_1), \dots, \phi(z_2) \rangle$ of Q_{2n-1}^1 . Therefore, we can construct n spanning paths of Q_{2n} , as illustrated in Figure 9.(b). Suppose s_1 and t_1 is on the same path. Without loss of generality, we can assume that one of these paths is $\langle s_2, \dots, y_1, s_1, \dots, t_1, y_2, \dots, t_2 \rangle$. There exists a Hamiltonian path between $\phi(y_1)$ and $\phi(y_2)$. Thus, we can construct n spanning paths of Q_{2n} .

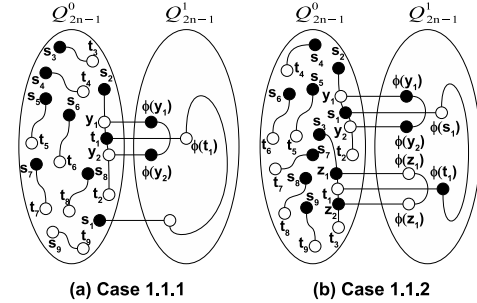


Figure 9: Illustration of Lemma 4

Case 1.2: $|K_{ww}^{01}| \geq 1$ or $|K_{ww}^{11}| \geq 1$.

We will use s_1 to denote a white vertex of K^{01} if $|K_{ww}^{01}| \geq 1$. When $|K_{ww}^{01}| = 0$, we use s_1 to denote a vertex of K_w^0 . Let $U_w^0 \subset (V_w^0 - K_w^0)$ with $|U_w^0| = |K_w^1| - 1$. Since $\{s_i, t_i\}_{\emptyset, K_b^0 \cup U_w^0 - \{s_1\}}$ is a connectable family, there exist $n - 1$ spanning paths of Q_{2n-1}^0 . Since $\{s_i, t_i\}_{\emptyset, \{\phi(s_1)\} \cup \phi(U_w^0)}$ is a connectable family, there exist $|K_w^1|$ spanning paths of

Q_{2n-1}^1 . Therefore, there exist n spanning paths of Q_{2n} , as illustrated in Figure 10.(a).

Case 1.3: $|K_{ww}^{01}| = 0$ and $|K_w^1| = 1$.

Let $t_1 \in K_w^1$ and $t_2 \in K_w^0$. Since $\{s_i, t_i\}_{\{t_2\}, K_b^0 - \{t_2\}}$ is a connectable family, there exist $\frac{|K^1|}{2} - 1$ spanning paths of Q_{2n-1}^0 . One of these paths is $\langle s_2, \dots, y, s_1 \rangle$. There exist two spanning paths $\langle \phi(y), \dots, \phi(t_2) \rangle$ and $\langle \phi(s_1), \dots, t_1 \rangle$ of Q_{2n-1}^1 . Therefore, there exist n spanning paths of Q_{2n} , as illustrated in Figure 10.(b).

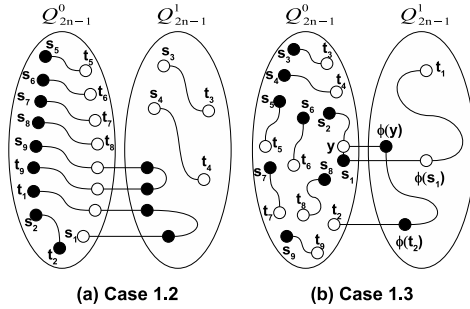


Figure 10: Illustration of **Lemma 4**

Case 1.4: $|K_{ww}^{01}| = 0$ and $|K_{ww}^{11}| \geq 1$ and $|K_w^1| \geq 2$.

Let $s_1, s_2 \in K_b^0$ and $y \in (V_w^0 - K_w^0)$ be a neighbor of s_1 . Let $U_w^0 \subset V_w^0 - K_w^0 - \{y\}$ with $|U_w^0| = |K_w^1| - 2$. Since $\{s_i, t_i\}_{\{y\}, K_b^0 \cup U_w^0}$ is a connectable family, there exist $n - 1$ spanning paths of Q_{2n-1}^0 . One of these paths is $\langle s_2, \dots, z, s_1 \rangle$. Since $\{s_i, t_i\}_{\{\phi(y), \phi(z)\} \cup \phi(U_w^0)}$ is a connectable family, there exist $|K_w^1|$ spanning paths of Q_{2n-1}^1 . Therefore, there exist n spanning paths of Q_{2n} , as illustrated in Figure 11.(a).

Case 2: $|K_b^0| \leq (n - 1)$ and $|K_w^0| \leq (n - 1)$.

Case 2.1: $|K^{11}| > 0$.

Let $U_b^0 \subset (V_b^0 - K_b^0)$ with $\phi(U_b^0) \subset (V_w^1 - K_w^1)$ and $|U_b^0| = |K_b^1| - 1$. Let $U_w^0 \subset (V_w^0 - K_w^0)$ with $\phi(U_w^0) \subset (V_b^1 - K_b^1)$ and $|U_w^0| = |K_w^1| - 1$. Since $\{s_i, t_i\}_{\{s_1\}, K_b^0 \cup U_b^0 - \{s_1\}}$ is a connectable family, there exist $n - 1$ spanning paths of Q_{2n-1}^0 . Since $\{s_i, t_i\}_{\{s_2\}, K_b^1 \cup \phi(U_w^0) - \{s_2\}}$ is a connectable family, there exist $|K^1| - 1$ spanning paths of Q_{2n-1}^1 . Therefore, there exist n spanning paths of Q_{2n} , as illustrated in Figure 11.(b).

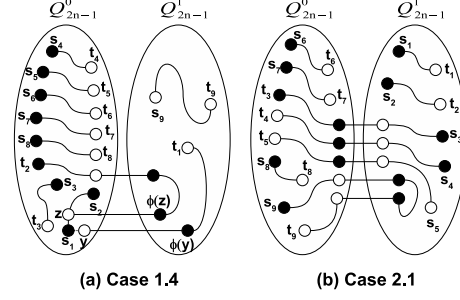


Figure 11: Illustration of **Lemma 4**

Case 2.2: $|K^{11}| = 0$ and $|K^0| \geq (n + 1)$ and $|K_{bb}| > 0$.

Without loss of generality, we can assume that $|K_w^{01}| \geq 1$. Let $s_1 \in K_w^{01}$. Let $U_b^0 \subset (V_b^0 - K_b^0)$ with $\phi(U_b^0) \subset (V_w^1 - K_w^1)$ and $|U_b^0| = |K_b^1|$. Let $U_w^0 \subset (V_w^0 - K_w^0)$ with $\phi(U_w^0) \subset (V_b^1 - K_b^1)$ and $|U_w^0| = |K_w^1| - 1$. Since $\{s_i, t_i\}_{\{s_1\}, K_b^0 \cup U_b^0 - \{s_1\}}$ is a connectable family, there exist $n - 1$ spanning paths of Q_{2n-1}^0 . Since $\{s_i, t_i\}_{\{\phi(s_1), \phi(t_2)\} \cup \phi(U_w^0)}$ is a connectable family, there exist $|K^1|$ spanning paths of Q_{2n-1}^1 . Therefore, there exist n spanning paths of Q_{2n} , as illustrated in Figure 12.(a).

Case 2.3: $|K^{11}| = 0$ and $(|K^0| = n \text{ or } |K_{bb}| = 0)$.

Without loss of generality, we can assume that $s_1 \in K_b^{01}$ and $s_2 \in K_b^{10}$. Let $U_b^0 \subset (V_b^0 - K_b^0)$ with $\phi(U_b^0) \subset (V_w^1 - K_w^1)$ and $|U_b^0| = |K_b^1| - 1$. Let $U_w^0 \subset (V_w^0 - K_w^0)$ with $\phi(U_w^0) \subset (V_b^1 - K_b^1)$ and $|U_w^0| = |K_w^1| - 1$. Since $\{s_i, t_i\}_{\{s_1\}, K_b^0 \cup U_b^0 - \{s_1\}}$ is a connectable family, there exist $n - 1$ spanning paths of Q_{2n-1}^0 . Since $\{s_i, t_i\}_{\{s_2\}, K_b^1 \cup \phi(U_w^0) - \{s_2\}}$ is a connectable family, there exist $|K^1| - 1$ spanning paths of Q_{2n-1}^1 . Therefore, there exist n spanning paths of Q_{2n} , as illustrated in Figure 12.(b). $\square$

By combining above two Lemma, we complete the result. Therefore, we draw the following theorem.

Theorem 1 Every balanced family $\{s_i, t_i\}_{F_b, K_b}^{F_w, K_w}$ of hypercube Q_n is connectable if $|F_b| + |F_w| + |K_b| + |K_w| \leq n, 4|F_b| + 2|K_b| \leq n + 1$ and $4|F_w| + 2|K_w| \leq n + 1$, for $n \geq 3$.

Proof: We will prove this theorem by induction on n . Since Q_3 is hyper Hamiltonian laceable, this

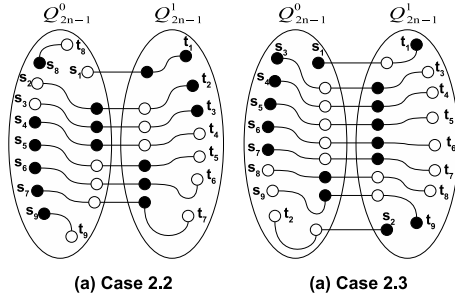


Figure 12: Illustration of Lemma 4

theorem is true for $n = 3$. Applying Lemma 3 and Lemma 4, we can prove the induction step. $\square$

Applying Theorem 1, we can obtain the following results.

Corollary 1 *There exists a fault-free path of $Q_n(V_b \cup V_w, E) - F_v$ with length $2^n - 2f_v - 1$ between two vertices with odd distance where $F_v \subset (V_b \cup V_w)$ and $f_v = \max\{|F_v \cap V_w|, |F_v \cap V_b|\} \leq \lceil \frac{n}{4} \rceil - 1$.*

Corollary 2 *The graph $Q_n(V_b \cup V_w, E) - F_v$ is Hamiltonian laceable for $|F_v \cap V_b| = |F_v \cap V_w| \leq \lceil \frac{n}{4} \rceil - 1$.*

Corollary 3 *There exists a fault-free cycle of $Q_n(V_b \cup V_w, E) - F_v$ with length $2^n - 2f_v$ where $F_v \subset (V_b \cup V_w)$ and $f_v = \max\{|F_v \cap V_w|, |F_v \cap V_b|\} \leq \lceil \frac{n}{4} \rceil - 1$.*

Corollary 4 *The graph $Q_n(V_b \cup V_w, E) - F_v$ is Hamiltonian for $|F_v \cap V_b| = |F_v \cap V_w| \leq \lceil \frac{n}{4} \rceil - 1$.*

Corollary 5 *There exists a fault-free path of $Q_n(V_b \cup V_w, E) - F_v$ with length $2^n - 2f_v$ between two vertices in V_i where $F_v \subset (V_b \cup V_w)$ and $f_v = \max\{|F_v \cap V_j|, |F_v \cap V_i| + 1\} \leq \lceil \frac{n-2}{4} \rceil$ for $i \neq j = b, w$.*

Corollary 6 *The graph $Q_n(V_b \cup V_w, E) - F_v$ is hyper-Hamiltonian laceable for $|F_v \cap V_b| = |F_v \cap V_w| \leq \lceil \frac{n-2}{4} \rceil - 1$.*

4 Conclusion

In this paper, we show that every balanced family $\{s_i, t_i\}_{F_b, K_b}^{F_w, K_w}$ of hypercube Q_n is connectable if $|F_b| + |F_w| + |K_b| + |K_w| \leq n$, $4|F_b| + 2|K_b| \leq n + 1$ and $4|F_w| + 2|K_w| \leq n + 1$, for $n \geq 3$. Applying this result, we can construct longer fault-free paths and cycles in hypercube with some vertex

faults. We also obtain some vertex fault tolerance Hamiltonian properties of hypercube. It is worthy to investigate similar properties in other bipartite interconnection networks, such as star networks, bubblesort networks.

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